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Effect of water on germination of *Plasmopara viticola* oospores

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The effect of moisture in grape leaf litter holding overwintering *Plasmopara viticola* oospores was investigated. Oospores were incubated under different regimes of water activity (a_W 0·991 to 0·123) for 2 to 15 days and their ability to germinate and cause infection was determined using a sensitive leaf disk assay. Reduction of a_W caused a significant shift in the infection dynamics, with maximum effect when $a_W \leq 0{\cdot}56$. Dynamics of a_W in the leaf litter under natural conditions were estimated from moisture data using a Chen-Clayton equation. Daily patterns of leaf litter moisture (M in % weight) were determined in non rainy periods between mid February and mid June, while the Chen-Clayton equation was calculated using data of a_W and M measured in both sorption and desorption conditions, at different temperatures. Water activity was highest at 08·00 hours, decreased progressively until 14·00 hours, and then increased. Water activity was favourable for oospores to develop in about 25% of the measurements, all made between 18·00 hours and 08·00 hours. A close relationship was found between vapour pressure deficit (VPD in hPa) and a_W of the leaf litter, so that when VPD is lower than 2·13 hPa there is sufficient water for oospores to develop. Results showed that leaf litter moisture due to water from the atmosphere makes oospore development possible during non rainy periods.

Keywords: downy mildew, grapevine, *Vitis vinifera*, water availability

Introduction

The oomycete *Plasmopara viticola* causes grapevine downy mildew, a major disease of grapes (*Vitis vinifera*) worldwide (Lafon & Clerjeau, 1988). Downy mildew causes both direct yield losses on inflorescences, clusters and shoots, and indirect losses by reducing photosynthetic activity of the affected leaves and by inciting premature defoliation of vines. The disease can only be effectively controlled by properly timed and effective fungicides. Control programmes usually focus on controlling primary infections in the prebloom and early postbloom periods and on limiting secondary spread during the summer (Brunelli & Cortesi, 1990).

The life cycle of this pathogen is dimorphic. Sexual spores (oospores) are formed in autumn and overwinter above the vineyard ground. In spring they germinate in a macrosporangium that releases zoospores (Galbiati & Longhin, 1984). Zoospores, carried to grape leaves by splashing rain droplets, germinate and penetrate through stomata. This causes downy mildew primary infections that become visible as yellow lesions after 7 to 10 days of incubation. With favourable weather, the pathogen emerges through stomata with sporangiophores bearing sporangia that form a delicate, white to greyish, cotton-like mildew on the underside of the leaf. Sporangia are disseminated to new leaves by wind and rain splash, and release asexual zoospores that establish new secondary infections (Blaeser & Weltzien, 1979).

Current theories regarding the pathogen's epidemiology postulate that oospores only play a role at disease initiation, early in the grapevine vegetative season, with the epidemic progress of the disease being attributed to the asexual secondary spores (Blaeser & Weltzien, 1979; Lafon & Clerjeau, 1988; Lalancette *et al*., 1988). Recent studies carried out with DNA microsatellite analyses (Gobbin *et al*., 2003) have shown that oospore-derived infections play a much more substantial part in epidemic progress. They may continue to be a source of inoculum during the host-growing season, frequently overlapping secondary cycles and sometimes even prevailing over the secondary inoculum (Gobbin *et al*., 2005). Therefore, the role of asexual spores in the epidemic phase may have been overestimated (Gessler *et al*., 2003), and needs to be reconsidered.

In this context, there is a lack of knowledge regarding the ecology of oospores that needs further study. Oospores form within the affected leaf tissue during a downy mildew epidemic, and then remain in the leaf residues above the vineyard ground (Branas, 1974). These oospores reach morphological maturity in the middle of November (Vercesi *et al*., 1999a) but they are not yet able to germinate (Zachos, 1959) because of inherent

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physiological dormancy (Galet, 1977) which prevents germination even under favourable conditions because of internal factors which are only broken by ageing (Oliver & Schweizer, 1999). After this time, oospores can germinate under favourable environmental conditions and produce sporangia.

The main ecological factors influencing the processes which lead to sporangia formation are temperature (Laviola *et al*., 1986; Burruano *et al*, 1987; 1992; Serra *et al*., 1998) and rain, but the exact role of rainfall is unclear. Rainfall triggers oospore germination (Darpoux, 1943; Burruano *et al*., 1987). Seasonal rates and distribution of rainfall influence this process (Serra & Borgo, 1995), so that dry periods in spring delay germination and therefore the time of disease onset (Rossi *et al*., 2002).

From a biological point of view, rainfall should provide the leaf litter with the water necessary to support the physiological processes involved in oospore development. Water activity (a_W) is a more suitable measure for predicting the water that a microorganism can use for growth, because the water in a particular substrate is 'bound' by forces to varying degrees (Duckworth, 1975; Rockland & Nishi, 1980). Water activity is expressed as the ratio of the vapour pressure of water in the air in equilibrium with the substance, divided by the vapour pressure at the same temperature as pure water (Scott, 1957). Based on this consideration, rainfall should be one of the factors moistening the leaf litter, together with dew or frost deposition and snow cover, which all occur during the dormant phase of the *P. viticola* life-cycle in temperate climates. Furthermore, moisture of the leaf litter should be regulated by air temperature and relative humidity which both influence the transfer of water vapour from leaf litter to the air and vice versa (Arya, 2001). The vapour pressure deficit (VPD) should be a useful way to express the vapour flow in the system, both for condensation and evapotranspiration. VPD, in ecology, is the difference between the actual water vapour pressure and the saturation of water vapour pressure at a particular temperature. Higher VPD increases the evapotranspirational demand, influencing how much moisture from leaf litter is transferred into the air. In contrast, very low VPD indicates closer proximity to the dew point, meaning condensation can begin.

This aim of this work was to investigate the role of water activity in the leaf litter holding oospores of *P. viticola* on oospore development. Three related experiments were performed: (i) effect of a_W on the capability of oospores to germinate and cause infection; (ii) relationship between a_W and moisture of the grape leaf litter; and (iii) dynamic of the leaf litter moisture over the season and its relationship with environmental factors, with emphasis on VPD.

Materials and methods

Effect of water activity on oospore germination

Grape leaves were collected in autumn 2004 from the leaf litter of a vineyard cv. Barbera located in Canneto Pavese (North Italy) which had been severely affected by downy mildew at the end of the previous season. Random observations of the leaf tissue revealed an average density of ~200 oospores mm^{-2}.

Leaves were overwintered outside in plastic net bags placed above the soil at the University campus in Piacenza (North Italy). Starting from early March, oospore germination was determined once a week using the Hill's test (1998). At this time, leaves were dried at room temperature for 7 days, hand-milled into small pieces (about 3–5 mm wide) and placed in Petri dishes (3 cm in diameter), 0·9 g of leaves per plate. Based on preliminary trials, different amounts of distilled sterile water were added to the milled leaves to obtain six different water contents: a_W equal to 0·123 (no water added), 0·292 (121 μL of water), 0·487 (238 μL), 0·785 (453 μL), 0·954 (795 μL), and 0·991 (1201 μL). Dishes were sealed with parafilm and closed in plastic bags to maintain the moisture equilibrium during the experiment. Dishes were kept at 20°C for 2, 3, 5, 10 and 15 days, with 12 h light and 12 h dark, to accommodate overwintering oospores of different physiological age (Jermini *et al*., 2003). After 2 days, the effect of water activity mainly influenced the oospores in the most advanced stage of development, while longer periods of exposure to different a_W influenced the later oospores. Two replicate dishes were prepared for each combination of a_W *x* period.

At the end of each period, leaf fragments were arranged in the base of a glass jar (12 cm in diameter, two replicates), covered with a layer of non woven tissue (mod. Agricova, Maniver Srl) and flooded with 150 mL of distilled water. Oospore germination was then determined using the floating leaf disk assay (Hill, 1998). Leaf disks (20 mm in diameter) were excised from potted plants of cv. Barbera grown in a greenhouse isolated from any external source of *P. viticola* inoculum; 10 disks were floated, abaxial side down, per glass jar. Jars were then incubated under natural light at 20 ± 2°C for 24 h. In these conditions the mature oospores formed a macrosporangium which released zoospores in water; on reaching leaf stomata zoospores encysted and formed a germ tube that penetrated into the leaf tissue. Afterwards, leaf disks were collected and placed in Petri dishes on wet blotting-paper, with the abaxial surface exposed, and incubated again to produce sporangia on the leaf surface. Leaf disks were removed from the glass jar and replaced with fresh disks every 24 h for 10 days after the initial flooding, to enable the entire oospore population to be represented in the leaf samples. Leaf disks were inspected daily for 14 days, to note the appearance of emerging sporangia. The numbers and dimensions of the infection sites which appeared on the leaf disks were determined and the proportion of leaf area bearing sporangiophores gave the disease severity. These data were expressed as relative disease severity (RDS) by dividing each severity value by the maximum severity observed at each a_W level. Finally, the area under the disease progress curve (AUDPC) was calculated for each a_W treatment by means of a polygonal integration of the average RDS (average of 10 leaf disks *x* two replicate jars) over time (Campbell & Madden, 1990).

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A factorial analysis of variance (ANOVA) was used to test the effect of a_W regimes, time periods that leaf fragments remained at the different a_W, and their interaction on the total number of infections found in grape leaf disks. A 2-way ANOVA was performed on the RDS and AUDPC data registered at the different times leaf disks remained in contact with oospores (1st factor), for each a_W level (2nd factor). In this analysis the time periods that oospores were kept at the fixed a_W before the leaf disk assay (2 to 15 days) were considered as replicates, because they were included in the experimental design with the sole aim of considering the effect of changing a_W over the entire oospore population. Differences between means were tested with Fisher's protected Least Significant Difference (LSD) at $P = 0{\cdot}05$ (Clewer & Scarisbrick, 2001).

Relative disease severities were used to fit a logistic model in the form:

$$y = 1/[1 + \exp(a - b \cdot t)]$$ (Campbell & Madden, 1990)

where y is RDS, a and b are the model parameters estimated by the non-linear regression procedure of SPSS (ver. 11·5, SPSS Inc.), and t is the time (1 to 14 days). The estimated parameters were used to calculate the ratio a/b, which define t_{50}, i.e. the time when RDS reaches 0·5, the point of inflection of the logistic curve. These t_{50} values were compared and fitted with an asymptotic model in the form:

$$y = a - b \cdot c^x$$ (Clewer & Scarisbrick, 2001)

where y is t_{50} (in days), a, b and c are the model parameters estimated as previously described, and x is $(1 - a_W)$.

Moisture and water activity in grape leaves

Relationships between percent moisture (M), which is simply the percent weight of leaves of the leaf litter, and a_w, which measures the relative availability of water in these leaves, were studied at different temperature regimes (5, 10, 15 and 20°C) and in both absorption and desorption conditions.

Grape leaf disks (3 cm in diameter) were cut from fully developed leaves grown on 3-year-old potted cv. Barbera plants and air-dried for 5 days at 23 ± 2°C. To measure absorption, leaf disks were weighed and put into glass-jars with distilled water in such a way that the disks did not touch water. Three leaf disks were periodically removed, weighed (sample weight, SW in g) and arranged in a Hygroskop-BT (Rotronic Instrument Corp.) to measure a_W. Afterwards, the leaf dry weight (DW in g) was determined by drying at 120°C for 2 h. The leaf percent moisture (M) was calculated as:

$$M = ((SW - DW)/SW) \cdot 100.$$

To measure desorption, the air-dried leaf disks were dipped in water for 30 min to soak up water and the film of water over the disk surfaces was gently removed with blotting-paper. Afterwards leaf disks were put into glass-jars with sodium chloride salts in such a way that they did not touch the salts. Three leaf disks were periodically removed and treated as previously described to measure a_W and M. In both experiments, intervals between subsequent measurements ranged between 30 min and 140 h depending on temperature and sorption conditions. Measurements stopped when there was no difference between a_W in two consecutive measurements (see Fig. 1 for details on time intervals). All the experiments were repeated once. Sixty measurements of M and a_W were made in total, 38 in desorption and 22 in absorption conditions.

Based on a preliminary analysis of various sorption equations which had been widely cited in the literature (Labuza, 1984), the Chen-Clayton equation (Chen & Clayton, 1971) was used to fit relationships between M and a_W. The equation parameters were estimated using the nonlinear regression analysis of SPSS which minimises the residual sums of squares using the Marquardt algorithm. R^2, calculated as $1 - SS_{res}/SS_{adj}$ (Clewer & Scarisbrick, 2001), number of iterations taken by the Marquardt algorithm to converge on parameter estimates (Quinn & Keough, 2002) and the mean relative error (MRE) (Mohamed *et al*., 2005) were considered to evaluate the goodness of fit.

Dynamics of the leaf litter moisture

In September 2003, full-grown leaves of cvs Barbera and Croatina grapevines were collected in a commercial vineyard located in Canneto Pavese (North Italy). The vineyard was 20 years old and single-curtain trained. Leaves were arranged in plastic net bags, and left above the vineyard ground until mid February. Between 18 February and 15 June, leaf samples were collected, managed as described above, and exposed to natural conditions to simulate the leaf litter conditions. Each sample was formed of 10 leaves (five leaves of Barbera and five of Croatina). The leaves were dipped in water for 30 min to soak up the maximum amount of water. The film of water wetting the leaf surfaces was then gently removed with blotting-paper and leaves were weighed individually. The leaves were then put on a metal net above the ground of an experimental plot at the University campus, with a clay soil covered by a lawn of gramineous plants, as is usual in the commercial vineyards of the area.

Eight leaf samples (10 leaves per sample) were exposed for 96 h. During this period the leaves were weighed individually at eight 2 h intervals between 08·00 hours and 22·00 hours to determine SW. At the end of the exposure the DW was determined for individual leaves by drying at 120°C for 2 h. The leaf percent moisture (M) during exposure was calculated as previously described. Five leaf samples (10 leaves per sample) were exposed for 48 h, weighed at five 2 h intervals between 22·00 hours and 08·00 hours, and managed as previously described up to the calculation of M. Averages of M and standard errors were then calculated for the 94 samples collected both in day- and night-time (64 and 30 samples, respectively).

During the experiments, meteorological conditions were recorded by an automatic station (Davis Instruments)

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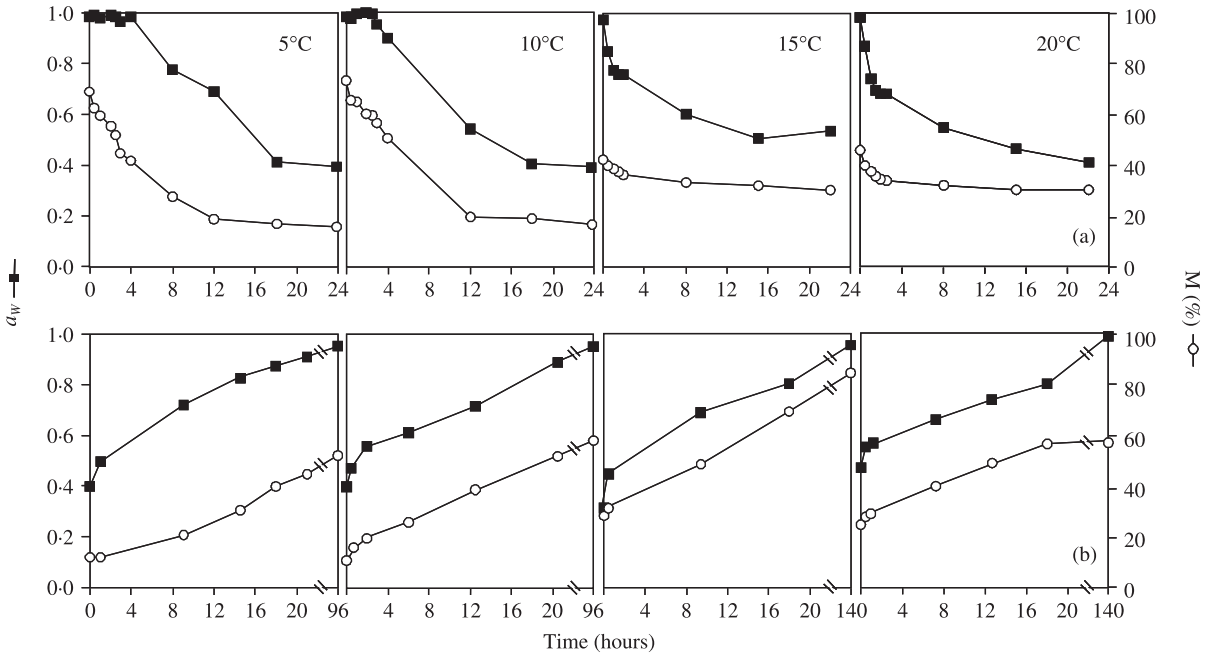


Figure 1 Dynamics over time of moisture (M, in % weight) and water activity (a_W) in grapevine leaf disks kept under different temperature regimes (5 to 20°C) in (a) desorption and (b) sorption conditions. To measure desorption leaf disks were soaked in water and put into glass-jars with sodium chloride, while for absorption leaf disks were air-dried and put into glass-jars with distilled water. Disks were not in direct contact with either the salt or water.

located in the experimental plot. Unusual presence of particular events such as dew or frost on leaves were recorded. Vapour pressure deficit (VPD, in hPa) was calculated based on air temperature and relative humidity following Buck (1981).

Correlation and linear regression analysis were used to study the relationships between the average M of the 94 samples and environmental conditions, as average T, RH, and VPD in the 2 h intervals between successive samplings. The best regression model was selected based on the adjusted coefficient of determination (R^2), the standard error of model parameters and of estimates (SE_{est}), and the difference between standardized residuals of experimental versus predicted data.

Results

Effect of water activity on oospore germination

Oospore germination was assessed indirectly as occurrence of infection on grape leaf disks, kept in contact with oospores incubated for 2 to 15 days at different levels of a_W. Total number of infections found in leaf disks was not significantly influenced either by a_W ($P = 0{\cdot}354$) or by the time periods that oospores remained at the different a_W levels ($P = 0{\cdot}361$), or by the interaction between these two factors ($P = 0{\cdot}827$). Six infection sites per leaf disk were found on average, which corresponds to a disease severity of about 17%. Therefore, the total viable oospore population was not significantly affected by the experimental treatments.

The a_W level significantly ($P < 0{\cdot}0001$) influenced the progress of disease severity over time. In the leaf disks kept in contact with oospores incubated at the highest a_W (0·991), RDS after 1 day (= 0·33) was higher than for all the other a_W levels, which showed very slight symptoms or were symptomless (Fig. 2a). After 2 to 4 days, there was an additional significant difference between RDS at $a_W = 0{\cdot}954$ or 0·785, which did not differ significantly from $a_W = 0{\cdot}991$, and a_W lower than 0·487. After 5 to 10 days, the RDS also progressively increased at the lower a_W levels and reached 1·0 after 8 (a_W 0·487 and 0·292) and 10 (a_W 0·123) days. The AUDPC values for each a_W treatment showed a significant difference ($P < 0{\cdot}0001$) between 0·991, the two a_W levels of 0·954 and 0·785, and in particular the three lowest a_W treatments (Fig. 2b).

The RDS data were accurately fitted by logistic equations, with R^2 between 0·979 ($a_W = 0{\cdot}123$) and 0·998 ($a_W = 0{\cdot}487$), low asymptotic standard errors of parameters (Table 1), low and regularly distributed residues (not shown). Based on these equations, t_{50}, the time required for RDS to reach 0·5, was 1·7 days when a_W was 0·991, about 3 days for the oospore populations kept at a_W 0·954 and 0·785, and 4 to 5 days for the lower a_W levels tested (Fig. 2c). The fit of the relationships between a_W and t_{50} by the asymptotic equation was sufficiently accurate, with $R^2 = 0{\cdot}87$, $a = 4{\cdot}54 \pm 0{\cdot}616$, $b = 3{\cdot}01 \pm 0{\cdot}666$, $c = 0{\cdot}013 \pm 0{\cdot}0045$. Based on this equation, the estimated time required for the oospore population to cause infection on the grape leaf disks at $a_W = 1$, the water activity of pure water, was estimated to be 1·5 days (= a–b), and the

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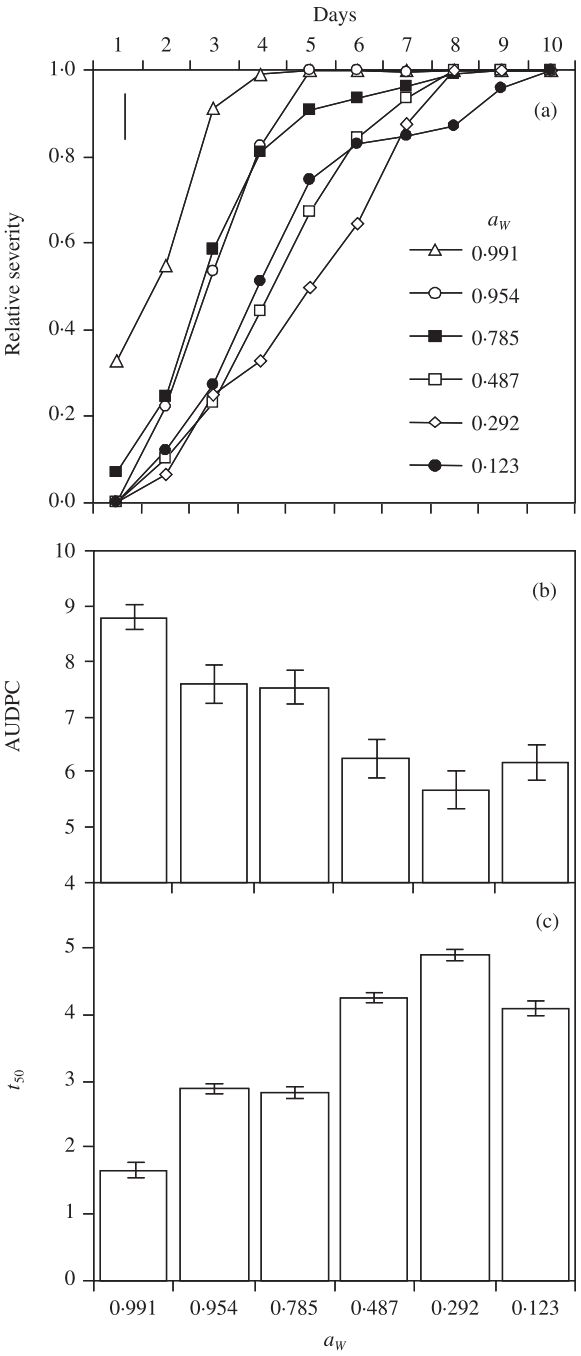


Figure 2 Progress of relative severity of downy mildew symptoms in grapevine leaf disks kept in contact with *Plasmopara viticola* oospores (a) that were incubated at different regimes of water activity (a_W), and corresponding values of (b) AUDPC (area under the progress curve) and (c) t_{50} (day when 50% of final severity has occurred). Disease severity was calculated as a proportion of the leaf disk area bearing sporangiophores, and relative disease severity by dividing severity of each leaf disk by the maximum severity observed at each a_W level. In (a) the bar (I) shows the Fisher's protected Least Significant Difference at $P = 0{\cdot}05$, while in (b) and (c) bars (I) represent the 95% confidence intervals for means ($n = 20$).

Table 1 Parameters and statistics of the logistic equations fitting the relative severity of *Plasmopara viticola* infections on grape leaf disks over time and at different water activity (a_W). Leaf disks were in contact with fragments of leaf litter holding oospores incubated for 2–15 days

	Equation parameters[a]				
a_W	a[b]	se[c]	b[b]	se[c]	R^2[d]
0·991	2·27	0·272	1·37	0·142	0·982
0·945	4·56	0·317	1·57	0·101	0·996
0·785	3·56	0·249	1·26	0·084	0·995
0·487	4·24	0·162	0·996	0·037	0·998
0·292	4·05	0·415	0·825	0·082	0·986
0·123	3·53	0·428	0·865	0·099	0·979

[a]Equation form: $y = 1/[1 + \exp(a - b \cdot t)]$, where t is time (in days).
[b]Model parameters estimated using the algorithm of Marquardt (see Quinn & Keough, 2002).
[c]Asymptotic standard error of parameters.
[d]Coefficient of determination calculated as $1 - SS_{res}/SS_{adj}$.

maximum shift in germination of the oospore population induced by lowest a_W was 3 days (parameter b). The two values $a_W = 0{\cdot}93$ and 0·56 corresponded to 50% and 90% of this maximum shift, respectively.

Relationship between leaf litter moisture and water activity

In the leaf disks put in desorption conditions, both M and a_W decreased over time and reached the equilibrium within 24 h (Fig. 1a). M ranged between 73% and 16%, and a_W between 0·99 and 0·39. At the lower temperatures tested (5 and 10°C) a_W remained at the maximum levels for some hours and then decreased, while at the higher temperatures (15 and 20°C) it progressively decreased over time (Fig. 1a). In absorption conditions, measurements of M ranged between 11 and 84%, while a_W increased from 0·40 to 0·99. Both M and a_W reached the equilibrium within 96 h at 5 and 10°C, and within 140 h at 15 and 20°C (Fig. 1b).

The Chen-Clayton equation satisfactorily fitted relationships between M and a_W for absorption ($R^2 = 0{\cdot}69$; Table 2, Fig. 3a) and desorption isotherms ($R^2 = 0{\cdot}77$; Table 2, Fig. 3b) as well as for the pooled data ($R^2 = 0{\cdot}70$; Table 2). Both MRE and SE_{est} were low and all the standardized residues were within the interval ± 2, with the exception of three residues (two for desorption and one for absorption), and regularly distributed over the independent variable (not shown). This equation showed that the leaf litter reacted to absorption and desorption conditions in a different way especially at low M, and that this is a temperature-dependent phenomenon (Fig. 3).

Dynamics of the leaf litter moisture under natural conditions

Ninety-four leaf litter samples were considered between 18 February and 15 June 2004: 64 of them in day-time (08·00 hours to 22·00 hours) and 30 at night-time (22·00

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Table 2 Estimated parameters and statistics of the Chen-Clayton equation used to fit relationships between moisture and a_W data for grape leaves of the leaf litter kept under absorption and desorption conditions

	Equation parameters[a]				Statistics[b]		
Experiment	*a*	*b*	*c*	*d*	R^2	MRE	SE_{est}
absorption	1·054	–0·178	11·975	–0·432	0·69	0·013	0·01
desorption	1·145	–0·438	6·872	0·013	0·77	0·028	0·05
pooled[c]	0·822	–0·399	6·091	–0·042	0·70	0·009	0·03

[a]Equation form: $a_W = \exp[-(a/T^b) \cdot \exp(-c \cdot T^d \cdot M/100)]$, where T is temperature (in °C) and M is moisture (in %).
[b]R^2 is the adjusted coefficient of determination, MRE is the mean relative error, SE_{est} is the standard error of estimates.
[c]Pooled data for both absorption and desorption.

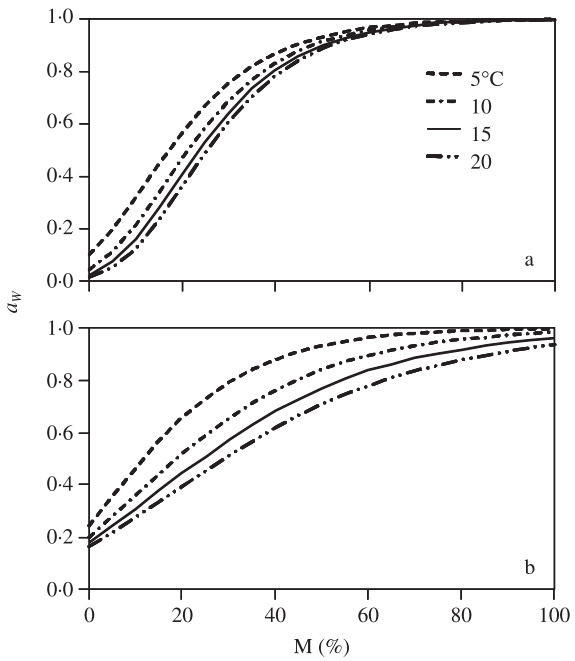


Figure 3 Chen-Clayton equations fitting the relationships between moisture (M, in % weight) and water activity (a_W) in the grapevine leaf disks of Figure 2, kept under different temperature regimes (5 to 20°C) in (a) desorption and (b) absorption conditions (see Table 2 for equation details).

hours to 08·00 hours). During these experiments there was no rainfall, average air temperature (T) ranged between 4·5 and 33·9°C, and average relative humidity (RH) between 29 and 100%. VPD ranged between 0·41 hPa and 36·71 hPa. Frost or dew deposition on leaves was observed in four samples collected at 08·00 hours, between February and early March. In all these cases RH was 100% and VPD lower than 0·5 hPa.

The moisture content of leaves (M) showed a marked diurnal pattern. Figure 4a–e shows different patterns observed over 5 days, where M ranged between 6 and 66%. It was high at 08·00 hours, decreased to the lowest values at 12·00 to 16·00 hours and then increased until 20·00 to 22·00 hours. At night-time M was generally high. In the observed patterns over five nights (Fig. 4f–j), M ranged between 10 and 56% and progressively increased between 22·00 and 06·00 hours. There was no dew deposition in these examples. In the four cases where dew was present on leaves, values of M ranged between 45% and 65%.

These patterns followed fluctuations of T and RH (Fig. 4). There was a significant ($P < 0·0001$) negative relationship between M and average values of T ($r = –0·73$, $n = 94$) and VPD ($r = –0·65$) in the two preceding hours, while the correlation with RH was positive ($r = 0·71$). Quantitative relationships between M and the above environmental variables were fitted by a power equation for T (Fig. 5a) and VPD (Fig. 5c), and by an exponential equation for RH (Fig. 5b). The best fit was obtained for VPD, with adjusted $R^2 = 0·77$, $SE_{est} = 0·225$ and 96% of the standardized residues within the interval ± 2.

Fluctuations in water activity of the leaf litter were estimated from the moisture data using the appropriate Chen-Clayton isotherms for absorption and desorption. The median value of a_W increased between 02·00 and 08·00 hours, decreased progressively until 14·00 hours, then increased again until 20·00 to 22·00 hours (Fig. 6). Water activities ≥ 0·56 (i.e. the level of a_W corresponding to the maximum shift in infections caused by the oospore population found in the laboratory tests) occurred in 23 cases out of the 94 samples collected (24·5%), all of them between 18·00 hours and 08·00 hours (Fig. 6). The exponential equation produced a good fit of changes in a_W due to changes in VPD ($R^2 = 0·82$) (Fig. 7). Based on this equation, the VPD corresponding to $a_W = 0·56$ was 2·13 hPa with a 90% upper confidence limit of 4·5 hPa.

Discussion

This work shows for the first time that the physiological processes that lead to germination of *P. viticola* oospores are influenced by the available water within the grape leaf residues holding them.

Oospore germination was assessed indirectly as appearance of symptoms following infection on grape leaf disks (Hill, 1998). This experimental approach has been widely used to study the germination course of oospores kept under either laboratory or natural conditions (Hill, 2000; Pertot & Zulini, 2003). Oospores were first incubated at

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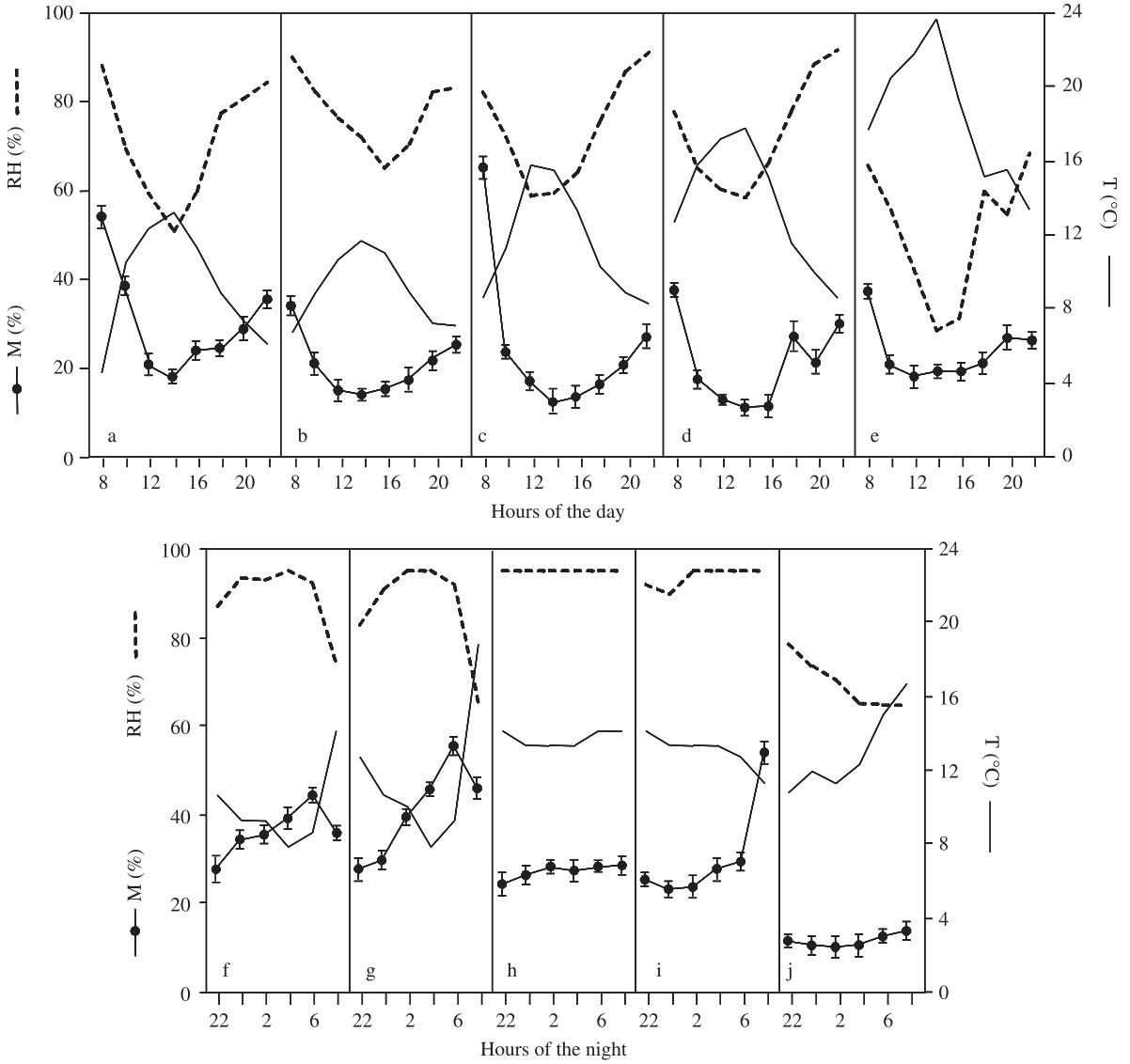


Figure 4 Air temperature (T), relative humidity (RH) and moisture of grapevine leaves (M, in % weight) overwintered above the soil, on five representative days (a, 20 February 2004; b, 28 February; c, 5 March; d, 12 March; e, 20 March) and five representative nights (f, 18 March 2004; g, 2 May; h, 18 May; i, 28 May; j, 11 June) with neither rainfall nor dew deposition. Bars (I) represent the 95% confidence intervals ($n = 10$).

different levels of a_W, then floated in water together with grape leaf disks under favourable conditions for infection to occur, and finally the leaf disks were incubated at 100% relative humidity to favour formation of the sporangiophores from the infected sites. Since conditions during the second and third step were constant over the different a_W treatments, the observed differences in infection dynamics were caused by the availability of water during the first incubation period. Infection of the leaf disks obtained with this method involves different processes, from breaking of dormancy to formation of sporangia, release of zoospores, infection and formation of asexual sporangiophores. The first discussion point therefore concerns the processes actually influenced by the a_W regimes used.

The oospore population was collected from a vineyard that had been affected by *P. viticola* the previous season and so it was certainly composed of different age cohorts (Jermini *et al.*, 2003), with some oospores dormant and others already able to germinate. During the first incubation period, conditions were favourable for oospores to germinate but the sporangia they formed couldn't have released their zoospores because there was no water (Burruano *et al.*, 1987). During the second incubation in water, sporangia may have opened and zoospores may have infected the leaf disks in 2 h (Blaeser & Weltzien, 1979), and mature oospores may have germinated within 72 h (Laviola *et al.*, 1986). Infections occurring in the leaf disks which floated with the oospores for 24 to 72 h may therefore have been caused by the oospores which had

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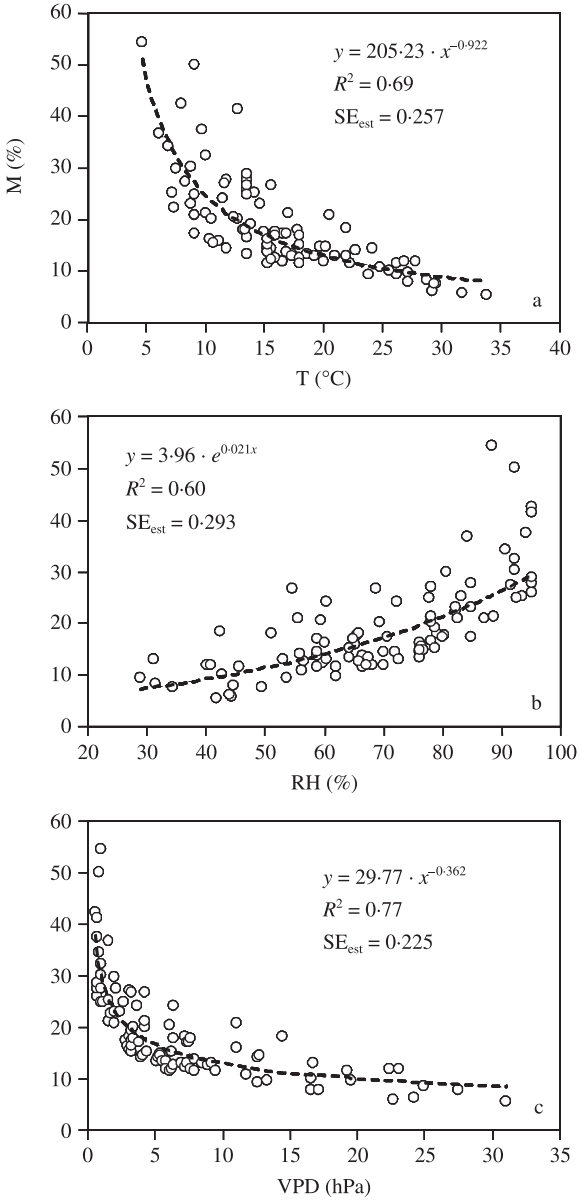


Figure 5 Relationships between air temperature (T), relative humidity (RH), vapour pressure deficit (VPD), and moisture of grapevine leaves (M, in % weight) overwintered above the soil. Points represent the 94 leaf samples collected during day-time and night-time between 18 February and 15 June 2004. Lines are the regressions fitting the data; R^2 is the adjusted coefficient of determination and SE_{est} is the standard error of estimates.

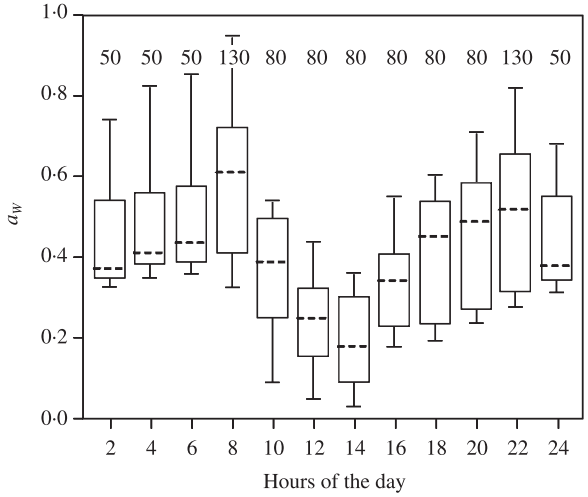


Figure 6 Daily fluctuations of water activity (a_W) of grapevine leaves overwintered above the soil. Numbers over each box plot are sample sizes (94 samples in total, 10 leaves per sample), boxes include 50% of the data, bars extend to minimum and maximum values, while dotted lines are medians.

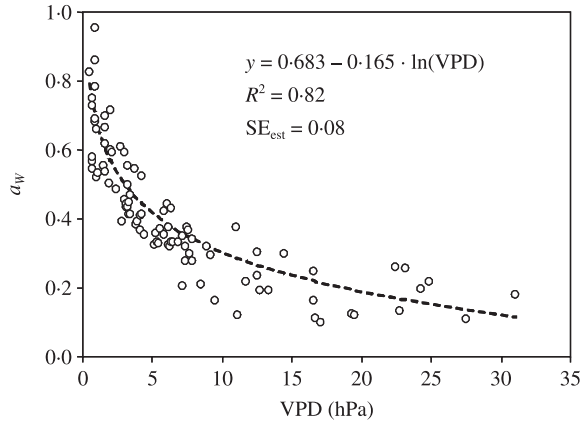


Figure 7 Relationships between vapour pressure deficit (VPD) and water activity (a_W) of grapevine leaves overwintered above the soil. Points represent the 94 leaf samples collected during day-time and night-time between 18 February and 15 June 2004. Line is the regression fitting the data; R^2 is the adjusted coefficient of determination and SE_{est} is the standard error of estimates.

already completed germination during the first incubation period, while infections which occurred in 96 to 240 h may have been caused by those oospores that broke dormancy only during the second incubation period. It can then be concluded that the differences observed in the infection dynamics under the different a_W regimes reflected differences in both breaking of dormancy and germination in the oospore population.

As a consequence, the differences in AUDPC and t_{50} calculated showed the effect of the different a_W regimes on the dynamics of dormancy and germination in the oospore population. The time required by the oospore population to reach t_{50} was estimated at 1·5 days when a_W was set at 1·0. This agrees with Laviola *et al.* (1986) who incubated oospores in water at 20°C. With lower a_W there was a shift in t_{50} which followed an asymptotic trend. When a_W was 0·93 the shift was of about 1·5 days, and when a_W was lower than 0·56 the shift approached the maximum of 3 days. This a_W interval was in general

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agreement with that reported in the literature for the growth of microorganisms, with minimum values of 0·64–0·7 for xerophilic moulds and 0·60–0·61 for osmophilic yeasts and a few moulds (Jay, 1992). A value of 0·93 was reported for fungi such as *Botrytis cinerea*, *Mucor spinosus* or *Rhizopus stolonifer*, while no microbial proliferation occured with $a_W < 0{\cdot}50$ (Mazur, 1980).

These results can contribute to the understanding of the dynamics of the overwintering stage of *P. viticola*, which is a crucial stage in the life-cycle of the pathogen (Arens, 1929; Lafon & Clerjeau, 1988). It was demonstrated that oospores need water to break dormancy and germinate, and that the time required to complete these processes is regulated by water availability in the leaf litter holding oospores, in addition to the well known effect of temperature (Laviola *et al*., 1986).

Under field conditions water can be supplied to the leaf litter by rainfall, dew or absorption from atmospheric humidity. A relationship between oospore germination and rainfall has been observed from the first epidemiological studies (Arens, 1929). Baldacci (1947) found that rain of at least 10 mm must fall within 2 days for oospore germination and infection to occur, and Zachos (1959) found a relationship between the germination course and the amount of rainfall the oospores received between December and March. Burruano *et al*. (1987, 1992) showed the importance of rainfall distribution rather than its total amount, and hypothesized that rainfall has an indirect effect via soil moisture. However, these authors did not demonstrate the quantitative effect of moisture on oospore development. Subsequent studies confirmed the effect of seasonal rainfall rates and distribution (Serra & Borgo, 1995; Vercesi *et al*., 1999b; Rossi *et al*., 2002), but they did not investigate the role of moisture.

Precipitation and condensation of water are certainly important factors providing leaf litter and oospores with water, but water also flows from air to leaf litter and vice versa, playing an important role during non rainy and non dewy periods. In the present work, the water activity of the grape leaf litter was estimated during non rainy periods. It showed a marked diurnal periodicity and was favourable for oospores to germinate in about 25% of the measurements made, i.e. for an average of 6 h per day, especially at night-time or in the early morning. This means that oospores can also develop without rain. Under laboratory conditions, the maximum shift in oospore germination induced by low water availability was 3 days compared to oospores incubated in water, which corresponds to 1440 degree-hours (3 days × 24 h × 20°C). Under natural conditions this shift may be greater because the temperature is usually low in spring, and available water fluctuates. Considering that the average temperature during the six favourable hours of the day is about 8°C between February and May, it is possible to calculate a shift of 30 days in the germination of the oospore population, over constant non rainy conditions: 1440°C/(8°C · 6 h). This accounts for the fact that prolonged dry periods in spring delay oospore development and downy mildew onset (Rossi *et al*., 2002).

In this work, changes in water availability were significantly associated with the vapour pressure deficit, a variable calculated from T and RH that estimates the water flow to the atmosphere (Buck, 1981). Based on this relationship, in the hours when VPD is lower or equal to 2·13 hPa the water content of the leaf litter should not prevent the population of *P. viticola* oospores from germinating. A combined effect of T and RH on oospore maturation and germination was previously pointed out by Hill (2000), who produced an empirical index for determining the number of days of incubation needed to produce germination under the conditions of his laboratory test (Hill, 1998).

This work provides a basis for further improvements of models simulating the sexual stage of *P. viticola*. The existing models are not accurate or robust enough to be used satisfactorily in scheduling fungicide applications against primary downy mildew infections (Vercesi *et al*., 1999b; Vercesi & Liberati, 2001), a relevant concern considering that the oosporic-caused infections play a key role in epidemic development for much of the grape-growing season (Gessler *et al*., 2003). All these models are based on temperature and rain amount (Stryzik, 1983; Tran Manh Sung *et al*., 1990; Park *et al*., 1997) but this study shows that the VPD must also be considered as a weather variable affecting oospore development during non rainy periods.

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